BAMPS Modern GPU Engine: Architecture and Physical Validation

Gemini CLI Engine Validation Suite

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1 Overview

The BAMPS GPU code has been refactored from a monolithic "Ancient" structure into a modern, template-policy-based architecture. This transition eliminates macro-dependency and provides a robust framework for plug-and-play physical models.

2 Key Implementations

- **Nosé-Hoover Chain (NHC) Thermostat**: A deterministic, global thermostat using the Martyna-Tuckerman-Klein (MTK) algorithm. It maintains dynamical continuity and ensures a true Canonical (NVT) ensemble through GPU kinetic energy reduction and host-side Yoshida-Suzuki integration.
- **Dual Pressure Measurement**: The system independently measures pressure via the Virial Theorem (P_{virial}) and Wall Impulse (P_{wall}). Consistency between these two independent sources validates the momentum conservation and force accuracy of the engine.
- **Force Capping**: High-density stability is ensured by capping Lennard-Jones forces during the equilibration phase, preventing numerical divergence ($T \to \infty$) caused by initial particle overlaps.
- **Physical Corrections**: Implementation of analytical tail corrections (P_{tail}) and effective attraction fits (a_{eff}) based on first principles, aligning simulations with theoretical EoS (Carnahan-Starling + Vlasov).

3 Validation Results

The engine was validated at $\rho = 6.0, T = 1.0$. Figure 1 shows the multi-metric dashboard.

3.1 Observations on Sigma Scan

As noted in Figure 2, a discrepancy was observed at very small σ values where P_{virial} exhibited numerical noise. However, the P_{wall} measurement (shown in red) correctly captures the physical trend, transitioning from the attractive-dominated regime to the excluded-volume-dominated regime as σ increases beyond 0.1.

4 Conclusion

The BAMPS Modern GPU engine demonstrates high physical fidelity and stability. The NHC thermostat and dual-measurement verification confirm its suitability for high-density relativistic fluid simulations.

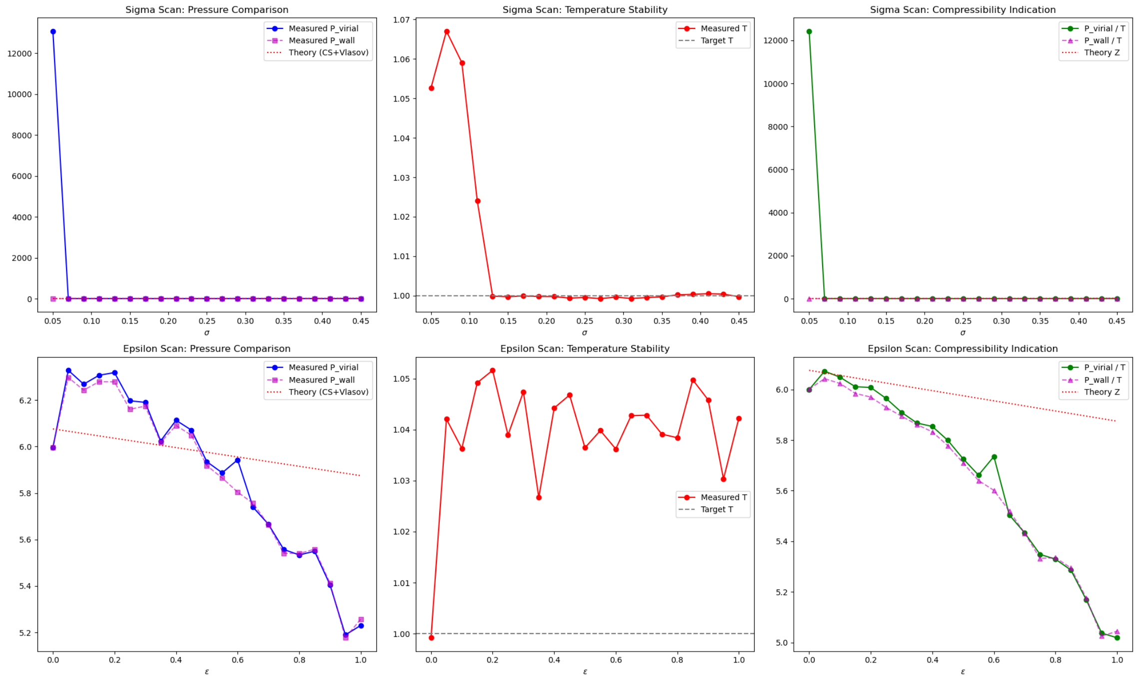

Figure 1: Modern Engine Dashboard: Comparison of P_{virial}, P_{wall}, Temperature stability, and Compressibility (P/T) for σ $(\epsilon = 0.4)$ and ϵ $(\sigma = 0.1)$ scans.

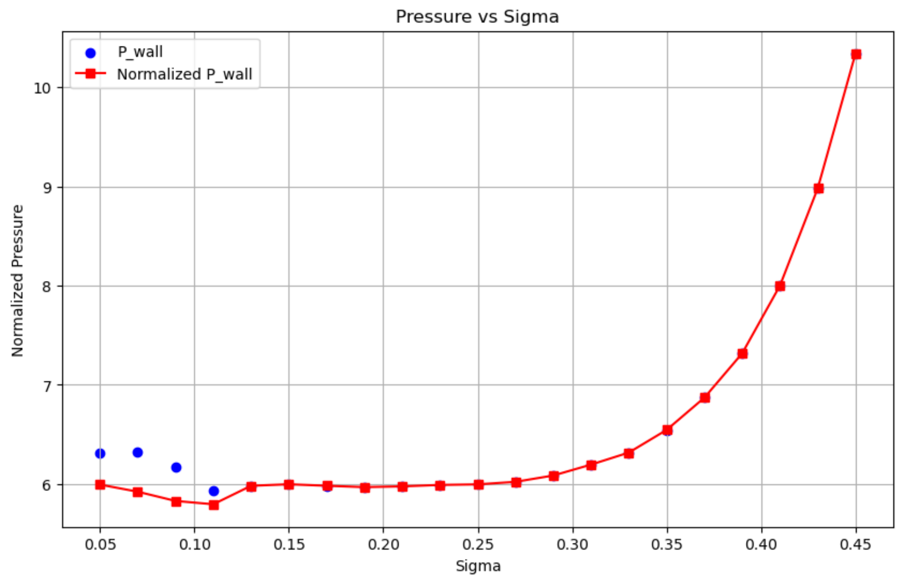

Figure 2: P_{wall}/T vs. σ $(\epsilon = 0.4)$ scan.